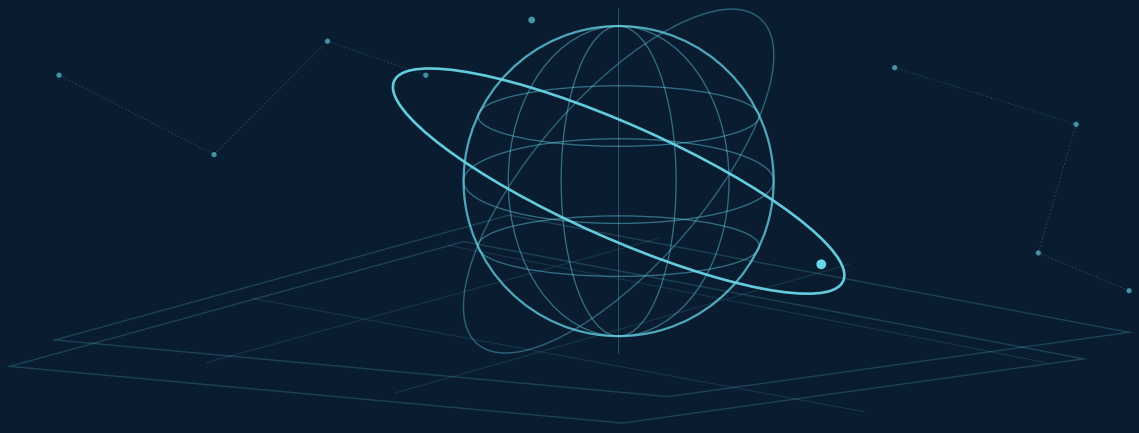


SPECIFICATION

Booking desk composition

Three use cases sharing an entity, a validation, a tool, and a quality requirement.

ATLAS-EX-COMPOSITION

Revision 2 / draft / Stable-v1.0.0

Document	ATLAS-EX-COMPOSITION	Revision	2
Status	draft	Language	en
Atlas version	Stable-v1.0.0	Profile version	Stable-v1.0.0
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Authors

Identity / name	Contract	Description / details
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Keywords: composition; booking; shared validation; case call

Context, actors, and scope

A booking desk creates, locates, and cancels bookings. The example is a specification, not an application.

The scope covers these three cases. Payment, customer management, and calendar integration are outside the example.

This is a synthetic specification example. Source SRC-BOOKING-BRIEF is a teaching baseline, not evidence of stakeholder approval or a claimed Needs-to-L1 derivation.

Actor

Identity / name	Contract	Description / details
ACT-BOOKING-CLERK Booking clerk		Kind: person Description: A person operating the booking desk.

Term

Identity / name	Contract	Description / details
VOC-BOOKING Booking		Definition: A retained reservation with a code, title, and lifecycle status. Synonyms: none

Entity and shared validations

The booking entity defines data, states, a transition, and two validations.

Type

Identity / name	Contract	Description / details
TYP-BOOKING-STATUS Booking status		Definition: The lifecycle state of a booking. Constraints: Known values are active and cancelled. Examples: active; cancelled

ENTITY ENT-BOOKING

Name Booking

Meaning

A booking records a named reservation and has a lifecycle status.

Identity

Description

Two bookings are the same when their codes are equal.

Fields: FLD-BOOKING-CODE

Fields

Identity / name	Contract	Description / details
FLD-BOOKING-CODE Booking code	Type: text Cardinality: 1 Required: true	Description: A stable uppercase code that identifies the booking. Constraints: 1 to 12 uppercase letters or digits. Examples: Valid: BOOK001; Invalid: book-001
FLD-BOOKING-TITLE Title	Type: text Cardinality: 1 Required: true	Description: A short human-readable title. Constraints: At least one non-whitespace character. Examples: Valid: Annual review; Invalid:
FLD-BOOKING-STATUS Status	Type: booking-status Cardinality: 1 Required: true	Description: The lifecycle state of the booking. Constraints: One of active or cancelled. Examples: Valid: active; cancelled; Invalid: deleted
FLD-BOOKING-VERSION Version	Type: integer Cardinality: 1 Required: true	Description: Concurrency token: zero for a new ready Booking, positive for a retained Booking. Constraints: A non-negative integer; successful save increments it. Examples: Valid: 0; 1; Invalid: -1

Relationships: none

States

Identity / name	Contract	Description / details
ST-BOOKING-ACTIVE Active		Description: The booking reserves its item.
ST-BOOKING-CANCELLED Cancelled		Description: The booking no longer reserves its item and is retained.

Transitions

Identity / name	Contract	Description / details
TRN-BOOKING-CANCEL Cancel	From: ST-BOOKING-ACTIVE To: ST-BOOKING-CANCELLED	Description: Cancellation changes status and retains the record.

Validations

VLD-BOOKING-CODE

Name Booking code format Scope field

Statement

The code is valid when it has 1 to 12 characters, all uppercase letters or digits.

Inputs

Identity / name	Contract	Description / details
Booking code	Source: field	Meaning: The value supplied for FLD-BOOKING-CODE.

Conditions

Identity / name	Contract	Description / details
C-VLD-BOOK-CODE-EMPTY The code is non-empty.		Meaning: An empty code fails.
C-VLD-BOOK-CODE-LEN The length is between 1 and 12.		Meaning: Longer codes fail.
C-VLD-BOOK-CODE-CHARS Every character is A-Z or 0-9.		Meaning: Lowercase and punctuation fail.

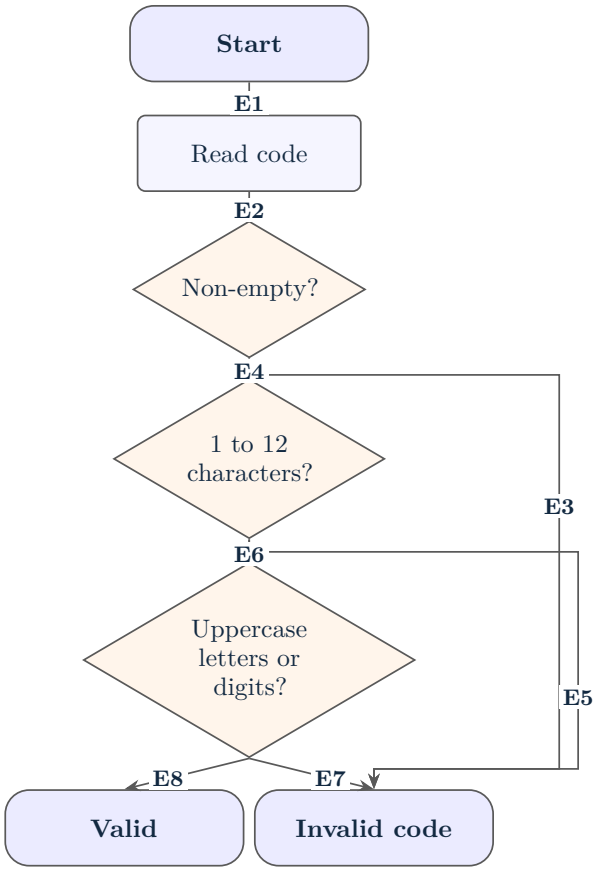
Results

Identity / name	Contract	Description / details
	Outcome: pass	Description: The code is accepted unchanged.
	Outcome: fail	Description: The code is rejected; the field keeps its previous value.

Flow GRAPH-VLD-BOOKING-CODE

Booking code validation

GRAPH-VLD-BOOKING-CODE



Key	Edge identity and label
E1	E-VLD-BOOK-CODE-1
E2	E-VLD-BOOK-CODE-2
E3	E-VLD-BOOK-CODE-3 — no
E4	E-VLD-BOOK-CODE-4 — yes
E5	E-VLD-BOOK-CODE-5 — no
E6	E-VLD-BOOK-CODE-6 — yes
E7	E-VLD-BOOK-CODE-7 — no
E8	E-VLD-BOOK-CODE-8 — yes

VLD-BOOKING-CANCEL

Name	Cancellation is allowed only from active	Scope	transition

Statement

A cancellation transition is valid only when the booking's current status is active.

Inputs

Identity / name	Contract	Description / details
Current status	Source: state	Meaning: The status read from the located booking.

Conditions

Identity / name	Contract	Description / details
C-VLD-BOOK-CANCEL-ACTIVE The current status is active.		Meaning: A cancelled booking fails this condition.

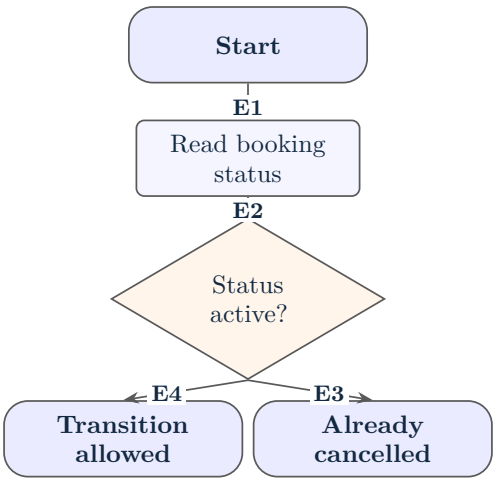
Results

Identity / name	Contract	Description / details
	Outcome: pass	Description: Cancellation may proceed.
	Outcome: fail	Description: The already-cancelled refusal applies; no save occurs.

Flow	GRAPH-VLD-BOOKING-CANCEL

Cancellation transition validation

GRAPH-VLD-BOOKING-CANCEL



Key	Edge identity and label
E1	E-VLD-BOOK-CANCEL-1
E2	E-VLD-BOOK-CANCEL-2
E3	E-VLD-BOOK-CANCEL-3 — no
E4	E-VLD-BOOK-CANCEL-4 — yes

`VLD-BOOKING-TITLE`

Name Booking title validity Scope field

Statement

The title is text with at least one non-whitespace character. Preserve valid text exactly.

Inputs

Identity / name	Contract	Description / details
Title	Source: field	Meaning: FLD-BOOKING-TITLE candidate.

Conditions

Identity / name	Contract	Description / details
C-VLD-BOOK-TITLE-TEXT The title has at least one non-whitespace character.		Meaning: Empty and whitespace-only titles fail.

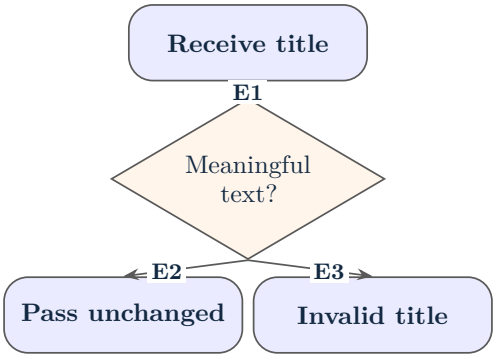
Results

Identity / name	Contract	Description / details
	Outcome: pass	Description: Accept unchanged.
	Outcome: fail	Description: Refuse before save.

Flow GRAPH-VLD-BOOKING-TITLE

Booking title validity

GRAPH-VLD-BOOKING-TITLE



Key	Edge identity and label
E1	E-VLD-BOOK-TITLE-1
E2	E-VLD-BOOK-TITLE-2 — yes
E3	E-VLD-BOOK-TITLE-3 — no

Examples

EX-BOOKING-VALID

Kind valid

Description

An active booking with a conforming code.

Data

```
{
"code": "BOOK001",
"title": "Annual review",
"status": "active",
"version": 0
}
```

EX-BOOKING-INVALID

Kind invalid

Description

A booking with a lowercase code and unknown status.

Data

```
{
"code": "book-001",
"title": "Annual review",
"status": "deleted",
"version": 0
}
```

Assumptions: none

Open issues: none

Shared abstract tool

One abstract tool serves all three cases through two operations.

TOOL TOL-BOOKING-STORE

Name Booking store

Purpose

Load and retain bookings. The contract is shared by creation, lookup, and cancellation and does not select a database.

Operations

OP-BOOKING-LOAD

Name Load booking

Purpose

Read one retained booking by code.

Inputs

Identity / name	Contract	Description / details
Booking code	Type: text Required: true	Meaning: The exact code to locate.

Conditions

The store is available.

Inputs described as ready entities already satisfy canonical entity invariants. Do not duplicate those validations inside this tool. When constructing entities from raw retained records, use the canonical entity hydration rules; operation-specific external conditions remain this tool's responsibility.

Results

Identity / name	Contract	Description / details
Booking	Type: BOOKING	Meaning: The retained booking with its current status. Guarantees: The result reflects the stored record at read time.

Effects

No persisted change.

Failures

Identity / name	Contract	Description / details
NOT_FOUND		Meaning: No retained booking has the code. Consumer distinction: Distinct from an invalid code and from an unavailable store.
STORE_UNAVAILABLE		Meaning: The store could not complete the read. Consumer distinction: Must not be reported as not found.

Guarantees

A successful load returns one current booking.

OP-BOOKING-SAVE

Name Save booking

Purpose

Persist the current data of one booking.

Inputs

Identity / name	Contract	Description / details
Booking	Type: BOOKING Required: true	Meaning: The booking to persist, including its status.
Expected version	Type: integer Required: true	Meaning: The version observed by the caller.

Conditions

The store is available.

Inputs described as ready entities already satisfy canonical entity invariants. Do not duplicate those validations inside this tool. When constructing entities from raw retained records, use the canonical entity hydration rules; operation-specific external conditions remain this tool's responsibility.

Results

Identity / name	Contract	Description / details
Saved booking	Type: BOOKING	Meaning: The booking with an incremented version. Guarantees: A successful save confirms the new data.

Effects

The retained booking is replaced with the supplied data.

Failures

Identity / name	Contract	Description / details
VERSION_CONFLICT		Meaning: The expected version is no longer current. Consumer distinction: This operation made no write. The existing or concurrently updated record is not overwritten.
STORE_UNAVAILABLE		Meaning: The store did not confirm the write. Consumer distinction: Distinguish write_outcome=not_applied, when supported by a no-write guarantee, from unknown. Do not report success or automatically retry an unknown write.

Guarantees

A successful save confirms one updated retained booking.

Expected version zero means insert only if the identifier is absent. A positive expected version means replace only the matching retained version. Comparison and write are atomic; success increments the version by one.

VERSION_CONFLICT guarantees this operation made no change; another concurrent operation may have changed the stored record.

Use cases and composition

Cancellation calls lookup. The response requirement is defined once and referenced by cancellation.

USE CASE UC-BOOKING-LOOKUP

Name Locate booking

Objective

Return the current data of one retained booking.

Actors

Actor ACT-BOOKING-CLERK Permission Permitted to locate bookings.

Description

The clerk operates the booking desk.

Resolve this permission from trusted context for the concrete initiating caller; deny absent or unresolved evidence.

Trigger

A clerk requests a booking by code.

Context

The lookup is used directly by the desk and by cancellation.

Caller context must contain a resolved authenticated principal, the relevant case permissions, and a validity status. Its origin is the application's trusted identity-and-permission authority; request fields and caller-supplied role strings are not trusted authorization evidence. Anonymous execution is not permitted. Missing, unresolved, expired or insufficient context is denied before any validation, tool access or effect.

The initiating actor is ACT-BOOKING-CLERK. The actor is a role; the caller is the concrete principal. The identity/authentication mechanism and channel encoding remain later design decisions.

Inputs

Identity / name	Contract	Description / details
Booking code	Type: text Required: true	Meaning: The code to locate.

Entry conditions

Requests may arrive without valid caller context; the explicit denial path handles them before any collaborator is invoked.

Entities

Entity ENT-BOOKING

Validations: VLD-BOOKING-CODE

Purpose

Validate the code before loading.

Tools

Tool	TOL-BOOKING-STORE	Operation	OP-BOOKING-LOAD

Purpose

Load the retained booking.

Calls: none

Outcomes

OUT-BOOKING-LOOKUP-FOUND

Kind success

Description

The booking is returned.

Effects

No persisted change.

OUT-BOOKING-LOOKUP-NOTFOUND

Kind refusal

Description

No booking has the code.

Effects

No persisted change.

OUT-BOOKING-LOOKUP-FAILURE

Kind failure

Description

The store is unavailable.

Effects

No persisted change.

OUT-BOOKING-LOOKUP-INVALID

Kind refusal

Description

The booking code is invalid.

Effects

No load or data change.

OUT-BOOKING-LOOKUP-DENIED

Kind refusal

Description

The trusted caller context does not authorize this case.

Effects

No validation, collaborator call, persisted change, or disclosure of booking existence.

Functional requirements

RF-BOOKING-LOOKUP-01

Statement

The case applies VLD-BOOKING-CODE and refuses an invalid code without loading.

Scope case

Applies to: UC-BOOKING-LOOKUP

Flow references: N-BOOK-LOOK-CODE, N-BOOK-LOOK-CODEC, N-BOOK-LOOK-INVALID, E-BOOK-LOOK-3

RF-BOOKING-LOOKUP-02

Statement

The case calls OP-BOOKING-LOAD and returns the loaded booking when found.

Scope operation

Applies to: UC-BOOKING-LOOKUP, OP-BOOKING-LOAD

Flow references: N-BOOK-LOOK-LOAD, N-BOOK-LOOK-FOUND, E-BOOK-LOOK-4, E-BOOK-LOOK-7

RF-BOOKING-LOOKUP-03

Statement

NOT_FOUND is reported as a distinct refusal with no effect.

Scope operation

Applies to: UC-BOOKING-LOOKUP, OP-BOOKING-LOAD

Flow references: N-BOOK-LOOK-NOTFOUND, E-BOOK-LOOK-5

RF-BOOKING-LOOKUP-04

Statement

STORE_UNAVAILABLE is reported as a failure, not as not found.

Scope operation

Applies to: UC-BOOKING-LOOKUP, OP-BOOKING-LOAD

Flow references: N-BOOK-LOOK-FAIL, E-BOOK-LOOK-6

RF-BOOKING-LOOKUP-AUTH

Statement

Require this case's permission in the trusted authenticated caller context. Deny missing, unresolved, expired or insufficient context before any validation, tool call, composed use-case call or effect. Anonymous execution is forbidden.

Scope case

Applies to: UC-BOOKING-LOOKUP

Flow references: N-BOOKING-LOOKUP-AUTH, N-BOOKING-LOOKUP-DENIED, E-BOOKING-LOOKUP-ALLOW, E-BOOKING-LOOKUP-DENY

Nonfunctional requirements: none

Flows

Main GRAPH-BOOKING-LOOKUP

Subflows: none

Flow text

Resolve and authorize the initiating caller first (RF-BOOKING-LOOKUP-AUTH). Denial terminates before any subsequent step.

Validate the code (RF-BOOKING-LOOKUP-01). An invalid code is refused before load.

Load through OP-BOOKING-LOAD (RF-BOOKING-LOOKUP-02). NOT_FOUND is a refusal (RF-BOOKING-LOOKUP-03); STORE_UNAVAILABLE is a failure (RF-BOOKING-LOOKUP-04).

Acceptance criteria

AC-BOOKING-LOOKUP-01

Statement

A retained code returns the booking and its current status.

Verifies: RF-BOOKING-LOOKUP-02

AC-BOOKING-LOOKUP-02

Statement

An unknown code returns not found, not an empty booking.

Verifies: RF-BOOKING-LOOKUP-03

AC-BOOKING-LOOKUP-03

Statement

An unavailable store is distinguished from not found.

Verifies: RF-BOOKING-LOOKUP-04

AC-BOOKING-LOOKUP-INVALID

Statement

A lowercase or punctuated code fails the canonical code rule and no load is invoked.

Verifies: RF-BOOKING-LOOKUP-01

AC-BOOKING-LOOKUP-AUTH

Statement

Missing, unresolved, expired or insufficient caller context produces denial with no validation, tool call, composed call or data disclosure.

Verifies: RF-BOOKING-LOOKUP-AUTH

Traceability

Identity / name	Contract	Description / details
TRC-BOOKING-LOOKUP-01	Source: RF-BOOKING-LOOKUP-01	Targets: VLD-BOOKING-CODE Note: Shared entity validation.
TRC-BOOKING-LOOKUP-ORIGIN	Source: SRC-BOOKING-BRIEF	Targets: RF-BOOKING-LOOKUP-01; RF-BOOKING-LOOKUP-02; RF-BOOKING-LOOKUP-03; RF-BOOKING-LOOKUP-04; RF-BOOKING-LOOKUP-AUTH Note: Versioned synthetic origin for these example obligations.

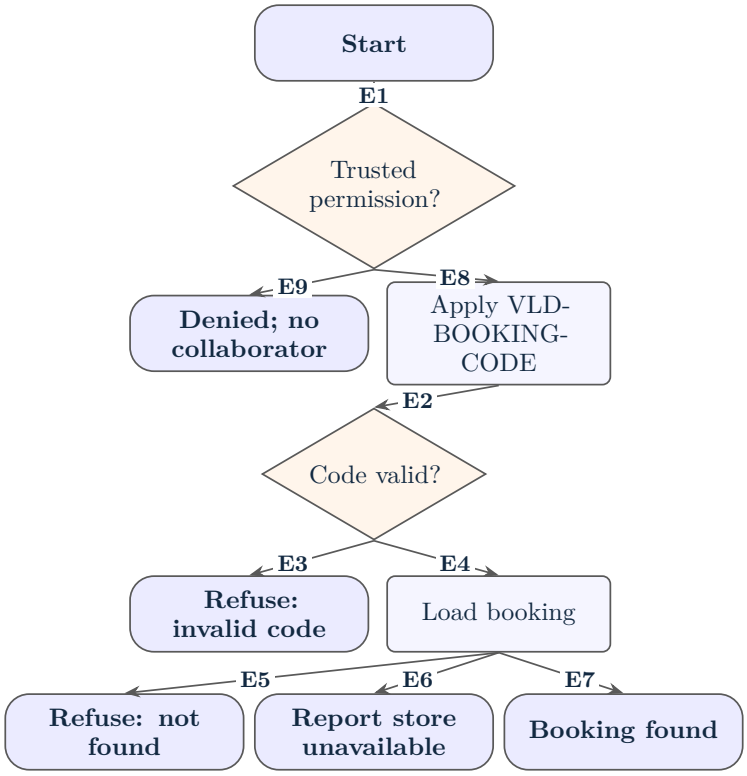
Assumptions: none

Sources: SRC-BOOKING-BRIEF

Open issues: none

Locate booking

GRAPH-BOOKING-LOOKUP



Key	Edge identity and label
E1	E-BOOK-LOOK-1
E2	E-BOOK-LOOK-2
E3	E-BOOK-LOOK-3 — fail
E4	E-BOOK-LOOK-4 — pass
E5	E-BOOK-LOOK-5 — NOT_FOUND
E6	E-BOOK-LOOK-6 — STORE_UNAVAILABLE
E7	E-BOOK-LOOK-7 — loaded

Key	Edge identity and label
E8	E-BOOKING-LOOKUP-ALLOW — authorized
E9	E-BOOKING-LOOKUP-DENY — missing / unresolved / insufficient

USE CASE `UC-BOOKING-CREATE`

Name Create booking

Objective

Create one active booking and confirm its stored state.

Actors

Actor ACT-BOOKING-CLERK Permission Permitted to create bookings.

Description

The clerk operates the booking desk.

Resolve this permission from trusted context for the concrete initiating caller; deny absent or unresolved evidence.

Trigger

A clerk supplies a code and title.

Context

The desk creates a booking in the shared store.

Caller context must contain a resolved authenticated principal, the relevant case permissions, and a validity status. Its origin is the application's trusted identity-and-permission authority; request fields and caller-supplied role strings are not trusted authorization evidence. Anonymous execution is not permitted. Missing, unresolved, expired or insufficient context is denied before any validation, tool access or effect.

The initiating actor is ACT-BOOKING-CLERK. The actor is a role; the caller is the concrete principal. The identity/authentication mechanism and channel encoding remain later design decisions.

Inputs

Identity / name	Contract	Description / details
Booking code	Type: text Required: true	Meaning: The code to create.
Title	Type: text Required: true	Meaning: The booking title.

Entry conditions

Requests may arrive without valid caller context; the explicit denial path handles them before any collaborator is invoked.

Entities

Entity ENT-BOOKING

Validations: VLD-BOOKING-CODE, VLD-BOOKING-TITLE

Purpose

Validate the code and form the active booking.

Tools

Tool	TOL-BOOKING-STORE	Operation	OP-BOOKING-SAVE

Purpose

Persist the new booking.

Calls: none

Outcomes

OUT-BOOKING-CREATE-DONE

Kind success

Description

The active booking is created.

Effects

One new booking is retained.

OUT-BOOKING-CREATE-INVALID

Kind refusal

Description

The code or title is invalid.

Effects

No save and no retained booking.

OUT-BOOKING-CREATE-CONFLICT

Kind failure

Description

The store reports a version conflict or unavailability.

Effects

No created booking is reported.

OUT-BOOKING-CREATE-DENIED

Kind refusal

Description

The trusted caller context does not authorize this case.

Effects

No validation, collaborator call, persisted change, or disclosure of booking existence.

Functional requirements

RF-BOOKING-CREATE-01

Statement

The case applies VLD-BOOKING-CODE and refuses an invalid code without saving.

Scope case

Applies to: UC-BOOKING-CREATE

Flow references: N-BOOK-CREATE-CODE, N-BOOK-CREATE-CODEC, N-BOOK-CREATE-INVALID, E-BOOK-CREATE-3

RF-BOOKING-CREATE-02

Statement

The case creates an active booking and calls OP-BOOKING-SAVE with version 0.

Scope operation

Applies to: UC-BOOKING-CREATE, OP-BOOKING-SAVE

Flow references: N-BOOK-CREATE-SAVE, E-BOOK-CREATE-4, E-BOOK-CREATE-7

RF-BOOKING-CREATE-03

Statement

VERSION_CONFLICT and STORE_UNAVAILABLE are reported as failures, never as a created booking.

Scope operation

Applies to: UC-BOOKING-CREATE, OP-BOOKING-SAVE

Flow references: N-BOOK-CREATE-DUP, N-BOOK-CREATE-FAIL, E-BOOK-CREATE-5, E-BOOK-CREATE-6

RF-BOOKING-CREATE-04

Statement

A successful result returns the created active booking.

Scope case

Applies to: UC-BOOKING-CREATE

Flow references: N-BOOK-CREATE-DONE, E-BOOK-CREATE-7

RF-BOOKING-CREATE-TITLE

Statement

After a valid code, apply VLD-BOOKING-TITLE. Refuse invalid title before creating a ready active Booking or invoking save.

Scope case

Applies to: UC-BOOKING-CREATE

Flow references: N-BOOK-CREATE-TITLE, N-BOOK-CREATE-TITLEVALID, E-BOOK-CREATE-TITLE-2, E-BOOK-CREATE-TITLE-3

RF-BOOKING-CREATE-AUTH

Statement

Require this case's permission in the trusted authenticated caller context. Deny missing, unresolved, expired or insufficient context before any validation, tool call, composed use-case call or effect. Anonymous execution is forbidden.

Scope case

Applies to: UC-BOOKING-CREATE

Flow references: N-BOOKING-CREATE-AUTH, N-BOOKING-CREATE-DENIED, E-BOOKING-CREATE-ALLOW, E-BOOKING-CREATE-DENY

Nonfunctional requirements

RNF-BOOKING-CREATE-RESPONSE

Scope interaction

Applies to: UC-BOOKING-CREATE

Statement

The boundary presents the result of a creation within 2 seconds of the stimulus for inputs within the declared limits.

Verifiable criterion

Visible result time is at most 2 seconds.

Verification

Run 30 creation requests against the declared configuration and record elapsed time.

Metric

Name	Visible result time	Unit	seconds
Target	2	Operator	<=

Measurement

Start

The creation stimulus is accepted.

End

The result is visible.

References: SRC-BOOKING-BRIEF

Applicability

Origin: SRC-BOOKING-BRIEF. This is a synthetic teaching obligation; any numeric target and scenario count are illustrative assumptions, not an accepted production SLA or a measurement made by this repository.

Flows

Main GRAPH-BOOKING-CREATE

Subflows: none

Flow text

Resolve and authorize the initiating caller first (RF-BOOKING-CREATE-AUTH). Denial terminates before any subsequent step.

Validate the code with VLD-BOOKING-CODE (RF-BOOKING-CREATE-01).

Apply title validity after code validity; refuse invalid title without saving (RF-BOOKING-CREATE-TITLE).

Create the active booking and save through OP-BOOKING-SAVE (RF-BOOKING-CREATE-02).

A save failure is reported as failure (RF-BOOKING-CREATE-03); a success returns the created booking (RF-BOOKING-CREATE-04).

Acceptance criteria

AC-BOOKING-CREATE-01

Statement

A valid code creates an active booking and a later lookup returns it.

Verifies: RF-BOOKING-CREATE-02, RF-BOOKING-CREATE-04

AC-BOOKING-CREATE-02

Statement

An invalid code is refused with no retained booking.

Verifies: RF-BOOKING-CREATE-01

AC-BOOKING-CREATE-03

Statement

A version conflict is never presented as a created booking.

Verifies: RF-BOOKING-CREATE-03

AC-BOOKING-CREATE-TITLE

Statement

A whitespace-only title is refused and OP-BOOKING-SAVE is not invoked.

Verifies: RF-BOOKING-CREATE-TITLE

AC-BOOKING-CREATE-AUTH

Statement

Missing, unresolved, expired or insufficient caller context produces denial with no validation, tool call, composed call or data disclosure.

Verifies: RF-BOOKING-CREATE-AUTH

AC-BOOKING-CREATE-QUALITY

Statement

In the declared didactic configuration, demonstrate the canonical quality criterion and its stated verification procedure; preserve its status as a teaching assumption, not a measured production claim.

Verifies: RNF-BOOKING-CREATE-RESPONSE

Traceability

Identity / name	Contract	Description / details
TRC-BOOKING-CREATE-01	Source: RF-BOOKING-CREATE-01	Targets: VLD-BOOKING-CODE Note: Shared validation also used by lookup.
TRC-BOOKING-CREATE-02	Source: RF-BOOKING-CREATE-02	Targets: OP-BOOKING-SAVE Note: Shared tool operation also used by cancellation.
TRC-BOOKING-CREATE-ORIGIN	Source: SRC-BOOKING-BRIEF	Targets: RF-BOOKING-CREATE-01; RF-BOOKING-CREATE-02; RF-BOOKING-CREATE-03; RF-BOOKING-CREATE-04; RF-BOOKING-CREATE-TITLE; RF-BOOKING-CREATE-AUTH; RNF-BOOKING-CREATE-RESPONSE Note: Versioned synthetic origin for these example obligations.

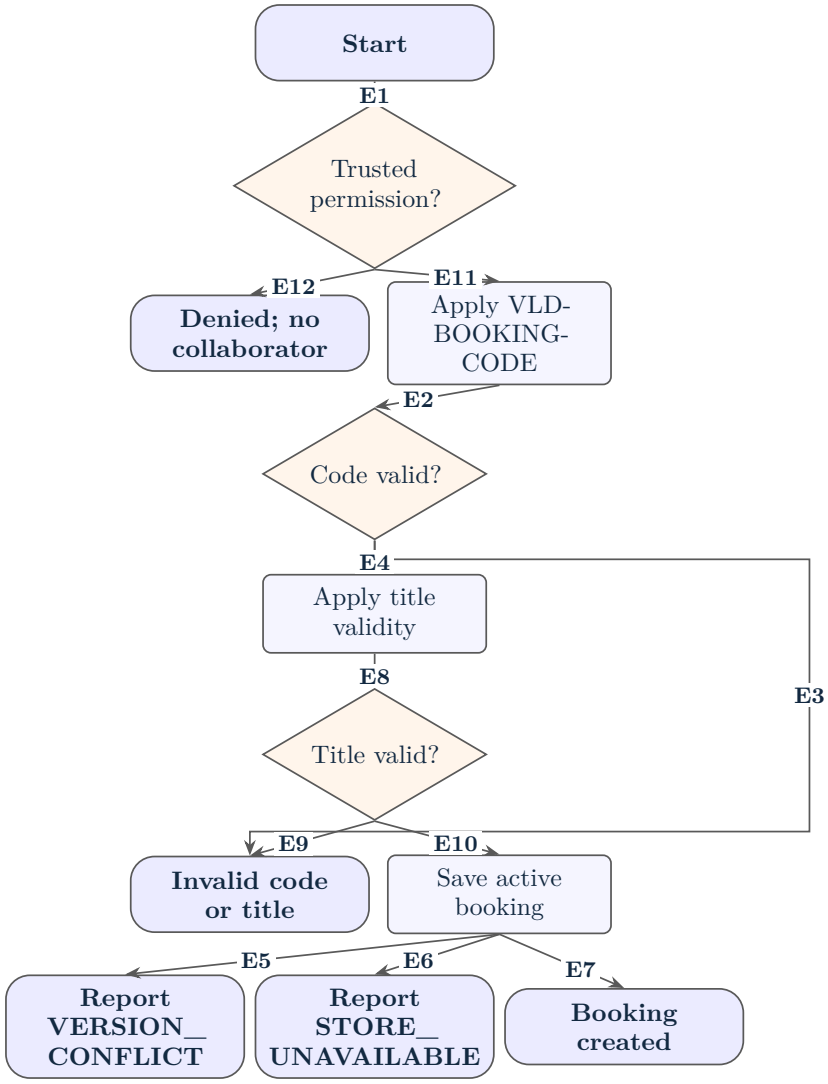
Assumptions: none

Sources: SRC-BOOKING-BRIEF

Open issues: none

Create booking

GRAPH-BOOKING-CREATE



Key	Edge identity and label
E1	E-BOOK-CREATE-1
E2	E-BOOK-CREATE-2
E3	E-BOOK-CREATE-3 — fail
E4	E-BOOK-CREATE-4 — pass
E5	E-BOOK-CREATE-5 — VERSION_CONFLICT
E6	E-BOOK-CREATE-6 — STORE_UNAVAILABLE
E7	E-BOOK-CREATE-7 — saved
E8	E-BOOK-CREATE-TITLE-1
E9	E-BOOK-CREATE-TITLE-2 — fail
E10	E-BOOK-CREATE-TITLE-3 — pass
E11	E-BOOKING-CREATE-ALLOW — authorized
E12	E-BOOKING-CREATE-DENY — missing / unresolved / insufficient

USE CASE UC-BOOKING-CANCEL

Name Cancel booking

Objective

Retain the booking and change its status to cancelled when cancellation is allowed.

Actors

Actor ACT-BOOKING-CLERK Permission Permitted to cancel bookings.

Description

The clerk operates the booking desk.

Resolve this permission from trusted context for the concrete initiating caller; deny absent or unresolved evidence.

Trigger

A clerk requests cancellation by code.

Context

Cancellation reuses lookup and the shared store.

Caller context must contain a resolved authenticated principal, the relevant case permissions, and a validity status. Its origin is the application's trusted identity-and-permission authority; request fields and caller-supplied role strings are not trusted authorization evidence. Anonymous execution is not permitted. Missing, unresolved, expired or insufficient context is denied before any validation, tool access or effect.

The initiating actor is ACT-BOOKING-CLERK. The actor is a role; the caller is the concrete principal. The identity/authentication mechanism and channel encoding remain later design decisions.

Inputs

Identity / name	Contract	Description / details
Booking code	Type: text Required: true	Meaning: The booking to cancel.

Entry conditions

Requests may arrive without valid caller context; the explicit denial path handles them before any collaborator is invoked.

Entities

Entity ENT-BOOKING

Validations: VLD-BOOKING-CANCEL

Purpose

Check that the located booking may move from active to cancelled.

Tools

Tool TOL-BOOKING-STORE Operation OP-BOOKING-SAVE

Purpose

Persist the cancelled status.

Calls

Use case UC-BOOKING-LOOKUP

Contract

Receives the located booking or the lookup refusal/failure.

Authorization

The cancellation actor must also be permitted to locate bookings.

Preserve the concrete principal and trusted caller context in the lookup invocation. Lookup independently checks its own permission; cancellation permission never bypasses it.

Outcomes

OUT-BOOKING-CANCEL-DONE

Kind success

Description

The booking is cancelled and retained.

Effects

The booking status becomes cancelled.

OUT-BOOKING-CANCEL-ALREADY

Kind refusal

Description

The booking is already cancelled.

Effects

No save and no data change.

OUT-BOOKING-CANCEL-NOTFOUND

Kind refusal

Description

The lookup did not find the booking.

Effects

No save and no data change.

OUT-BOOKING-CANCEL-FAILURE

Kind failure

Description

The store or lookup failed.

Effects

No cancellation is claimed.

OUT-BOOKING-CANCEL-INVALID

Kind refusal

Description

Lookup refused an invalid code.

Effects

No save or data change.

OUT-BOOKING-CANCEL-DENIED

Kind refusal

Description

The trusted caller context does not authorize this case.

Effects

No validation, collaborator call, persisted change, or disclosure of booking existence.

Functional requirements

RF-BOOKING-CANCEL-01

Statement

The case calls UC-BOOKING-LOOKUP and preserves its not-found and invalid-code refusals.

Preserve the caller context and propagate lookup denial before any save.

Scope case

Applies to: UC-BOOKING-CANCEL

Flow references: N-BOOK-CANCEL-LOOK, N-BOOK-CANCEL-FOUND, N-BOOK-CANCEL-NOTFOUND, N-BOOK-CANCEL-INVALID, N-BOOK-CANCEL-LOOKDENIED, E-BOOK-CANCEL-3, E-BOOK-CANCEL-4, E-BOOK-CANCEL-INVALID, E-BOOK-CANCEL-LOOKDENIED, N-BOOK-REFUSE-NOTFOUND, E-BOOK-REFUSE-2

RF-BOOKING-CANCEL-02

Statement

The case applies VLD-BOOKING-CANCEL to the located booking before any save.

Scope case

Applies to: UC-BOOKING-CANCEL

Flow references: N-BOOK-CANCEL-VALIDATE, N-BOOK-CANCEL-VALID, E-BOOK-CANCEL-5

RF-BOOKING-CANCEL-03

Statement

An already-cancelled booking produces the already-cancelled refusal with no save and no effect.

Scope case

Applies to: UC-BOOKING-CANCEL

Flow references: N-BOOK-CANCEL-ALREADY, E-BOOK-CANCEL-6, N-BOOK-REFUSE-START, N-BOOK-REFUSE-ALREADY, E-BOOK-REFUSE-1

RF-BOOKING-CANCEL-04

Statement

A passing transition sets status cancelled and calls OP-BOOKING-SAVE with the observed version.

Scope operation

Applies to: UC-BOOKING-CANCEL, OP-BOOKING-SAVE

Flow references: N-BOOK-CANCEL-SAVE, N-BOOK-CANCEL-DONE, E-BOOK-CANCEL-7, E-BOOK-CANCEL-9

RF-BOOKING-CANCEL-05

Statement

A store failure is reported distinctly and does not claim cancellation.

Scope operation

Applies to: UC-BOOKING-CANCEL, OP-BOOKING-SAVE

Flow references: N-BOOK-CANCEL-ERROR, E-BOOK-CANCEL-8, E-BOOK-CANCEL-LOOKERROR

RF-BOOKING-CANCEL-AUTH

Statement

Require this case's permission in the trusted authenticated caller context. Deny missing, unresolved, expired or insufficient context before any validation, tool call, composed use-case call or effect. Anonymous execution is forbidden.

Scope case

Applies to: UC-BOOKING-CANCEL

Flow references: N-BOOKING-CANCEL-AUTH, N-BOOKING-CANCEL-DENIED, E-BOOKING-CANCEL-ALLOW, E-BOOKING-CANCEL-DENY

Nonfunctional requirements

RNF-BOOKING-CANCEL-RESPONSE

Reference to RNF-BOOKING-CREATE-RESPONSE

Scope interaction

Applies to: UC-BOOKING-CANCEL

References: SRC-BOOKING-BRIEF

Applicability

Origin: SRC-BOOKING-BRIEF. This is a synthetic teaching obligation; any numeric target and scenario count are illustrative assumptions, not an accepted production SLA or a measurement made by this repository.

Flows

Main GRAPH-BOOKING-CANCEL

Subflows: GRAPH-BOOKING-CANCEL-REFUSAL

Flow text

Resolve and authorize the initiating caller first (RF-BOOKING-CANCEL-AUTH). Denial terminates before any subsequent step.

Call UC-BOOKING-LOOKUP (RF-BOOKING-CANCEL-01). Its refusals and failures remain observable.

Apply VLD-BOOKING-CANCEL to the located booking (RF-BOOKING-CANCEL-02).

An already-cancelled booking reaches the refusal subflow (RF-BOOKING-CANCEL-03) with no save.

On pass, set status cancelled and save with the observed version (RF-BOOKING-CANCEL-04). A save failure is reported (RF-BOOKING-CANCEL-05).

A composed lookup denial terminates cancellation before save. A lookup failure is propagated as failure, never collapsed into not found.

Acceptance criteria

AC-BOOKING-CANCEL-01

Statement

Cancelling an active booking returns cancelled and retains the code.

Verifies: RF-BOOKING-CANCEL-02, RF-BOOKING-CANCEL-04

AC-BOOKING-CANCEL-02

Statement

Cancelling an already-cancelled booking is refused with no data change.

Verifies: RF-BOOKING-CANCEL-03

AC-BOOKING-CANCEL-03

Statement

A lookup not-found is distinguishable from an already-cancelled refusal.

Verifies: RF-BOOKING-CANCEL-01, RF-BOOKING-CANCEL-03

AC-BOOKING-CANCEL-FAILURE

Statement

An unavailable lookup or save remains a failure; an unknown write is not reported as no change, success, or retried automatically.

Verifies: RF-BOOKING-CANCEL-05

AC-BOOKING-CANCEL-CONTEXT

Statement

A caller allowed to cancel but not to locate is denied by the composed lookup, with the same principal preserved and no save.

Verifies: RF-BOOKING-CANCEL-01

AC-BOOKING-CANCEL-AUTH

Statement

Missing, unresolved, expired or insufficient caller context produces denial with no validation, tool call, composed call or data disclosure.

Verifies: RF-BOOKING-CANCEL-AUTH

AC-BOOKING-CANCEL-QUALITY

Statement

In the declared didactic configuration, demonstrate the canonical quality criterion and its stated verification procedure; preserve its status as a teaching assumption, not a measured production claim.

Verifies: RNF-BOOKING-CANCEL-RESPONSE

Traceability

Identity / name	Contract	Description / details
TRC-BOOKING-CANCEL-01	Source: RF-BOOKING-CANCEL-02	Targets: VLD-BOOKING-CANCEL; TRN-BOOKING-CANCEL Note: Transition validation and entity transition.
TRC-BOOKING-CANCEL-02	Source: RNF-BOOKING-CANCEL-RESPONSE	Targets: RNF-BOOKING-CREATE-RESPONSE Note: Explicit shared obligation reference.
TRC-BOOKING-CANCEL-ORIGIN	Source: SRC-BOOKING-BRIEF	Targets: RF-BOOKING-CANCEL-01; RF-BOOKING-CANCEL-02; RF-BOOKING-CANCEL-03; RF-BOOKING-CANCEL-04; RF-BOOKING-CANCEL-05; RF-BOOKING-CANCEL-AUTH; RNF-BOOKING-CANCEL-RESPONSE Note: Versioned synthetic origin for these example obligations.

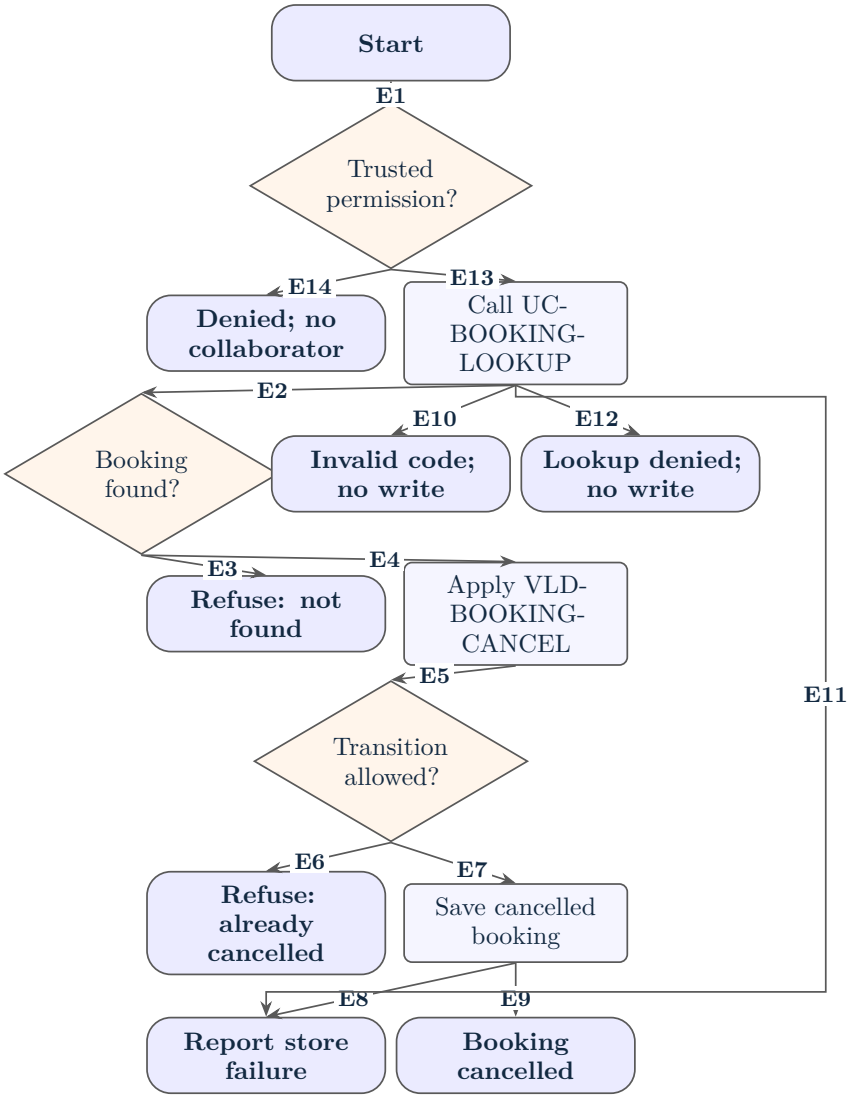
Assumptions: none

Sources: SRC-BOOKING-BRIEF

Open issues: none

Cancel booking

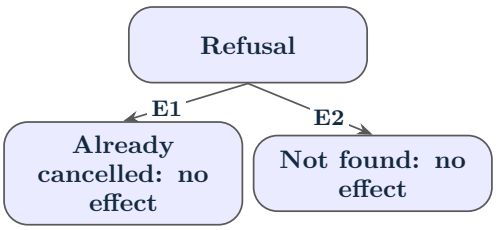
GRAPH-BOOKING-CANCEL



Key	Edge identity and label
E1	E-BOOK-CANCEL-1
E2	E-BOOK-CANCEL-2 — found / not found
E3	E-BOOK-CANCEL-3 — no
E4	E-BOOK-CANCEL-4 — yes
E5	E-BOOK-CANCEL-5
E6	E-BOOK-CANCEL-6 — fail
E7	E-BOOK-CANCEL-7 — pass
E8	E-BOOK-CANCEL-8 — failure
E9	E-BOOK-CANCEL-9 — saved
E10	E-BOOK-CANCEL-INVALID — invalid code
E11	E-BOOK-CANCEL-LOOKERROR — lookup failure
E12	E-BOOK-CANCEL-LOOKDENIED — lookup denied
E13	E-BOOKING-CANCEL-ALLOW — authorized
E14	E-BOOKING-CANCEL-DENY — missing / unresolved / insufficient

Cancellation refusal subflow

GRAPH-BOOKING-CANCEL-REFUSAL



Key	Edge identity and label
E1	E-BOOK-REFUSE-1 — already cancelled
E2	E-BOOK-REFUSE-2 — not found

Interaction boundary

The command line exposes all three cases and maps their observable outcomes.

BOUNDARY BND-BOOKING-CLI

Name	Booking desk command line	Modality	cli

Purpose

Expose booking creation, lookup, and cancellation through commands.

Origin classification: this channel is an illustrative solution proposal. Its form, command syntax, layout and encoding are outside the L1 conformance claim. The referenced cases define domain inputs and outcomes independently of this channel.

Audience

Desk clerks operating a local terminal.

Interactions

INT-BOOKING-CREATE

Name	Create command

Description

Accepts a code and title and presents the created booking or a refusal/failure.

Illustration only; no channel-specific UI or command choice is promoted to a domain RF by appearing here.

Use case	UC-BOOKING-CREATE

Trigger

The clerk invokes booking create.

Input mapping

Source	argv.code	Target	UC-BOOKING-CREATE.input.code

Meaning

The booking code argument.

Source	argv.title	Target	UC-BOOKING-CREATE.input.title

Meaning

The title argument.

Result mapping

Source	OUT-BOOKING-CREATE-DONE	Target	stdout.created

Meaning

The active booking is created.

Source	OUT-BOOKING-CREATE-INVALID	Target	stderr.invalid

Meaning

The code or title is invalid.

Source	OUT-BOOKING-CREATE-CONFLICT	Target	stderr.failure

Meaning

The store reports a version conflict or unavailability.

Source	OUT-BOOKING-CREATE-DENIED	Target	result.out-booking-create-denied

Meaning

The trusted caller context does not authorize this case.

Requirement refs: RF-BOOKING-CREATE-01, RF-BOOKING-CREATE-02, RF-BOOKING-CREATE-03, RF-BOOKING-CREATE-04, RF-BOOKING-CREATE-TITLE, RF-BOOKING-CREATE-AUTH

Resources: none

Examples

```
booking create BOOK001 "Annual review"
```

INT-BOOKING-CANCEL

Name Cancel command

Description

Accepts a code and presents cancellation, already-cancelled, not-found, or failure.

Illustration only; no channel-specific UI or command choice is promoted to a domain RF by appearing here.

Use case UC-BOOKING-CANCEL

Trigger

The clerk invokes booking cancel.

Input mapping

Source	argv.code	Target	UC-BOOKING-CANCEL.input.code

Meaning

The booking code argument.

Result mapping

Source	OUT-BOOKING-CANCEL-DONE	Target	stdout.cancelled

Meaning

The booking is cancelled and retained.

Source	OUT-BOOKING-CANCEL-ALREADY	Target	stderr.already

Meaning

The booking is already cancelled.

Source	OUT-BOOKING-CANCEL-NOTFOUND	Target	stderr.not_found

Meaning

The lookup did not find the booking.

Source	OUT-BOOKING-CANCEL-FAILURE	Target	stderr.failure

Meaning

The store or lookup failed.

Source	OUT-BOOKING-CANCEL-INVALID	Target	result.out-booking-cancel-invalid

Meaning

Lookup refused an invalid code.

Source	OUT-BOOKING-CANCEL-DENIED	Target	result.out-booking-cancel-denied

Meaning

The trusted caller context does not authorize this case.

Requirement refs: RF-BOOKING-CANCEL-01, RF-BOOKING-CANCEL-02, RF-BOOKING-CANCEL-03, RF-BOOKING-CANCEL-04, RF-BOOKING-CANCEL-05, RF-BOOKING-CANCEL-AUTH

Resources: none

Examples

```
booking cancel BOOK001
```

INT-BOOKING-LOOKUP

Name Lookup command

Description

Accepts a code and presents the booking, not found, or failure.

Illustration only; no channel-specific UI or command choice is promoted to a domain RF by appearing here.

Use case UC-BOOKING-LOOKUP

Trigger

The clerk invokes booking lookup.

Input mapping

Source	argv.code	Target	UC-BOOKING-LOOKUP.input.code

Meaning

The booking code argument.

Result mapping

Source	OUT-BOOKING-LOOKUP-FOUND	Target	stdout.booking

Meaning

The booking is returned.

Source	OUT-BOOKING-LOOKUP-NOTFOUND	Target	stderr.not_found

Meaning

No booking has the code.

Source	OUT-BOOKING-LOOKUP-FAILURE	Target	stderr.failure

Meaning

The store is unavailable.

Source	OUT-BOOKING-LOOKUP-INVALID	Target	result.out-booking-lookup-invalid

Meaning

The booking code is invalid.

Source	OUT-BOOKING-LOOKUP-DENIED	Target	result.out-booking-lookup-denied

Meaning

The trusted caller context does not authorize this case.

Requirement refs: RF-BOOKING-LOOKUP-01, RF-BOOKING-LOOKUP-02, RF-BOOKING-LOOKUP-03, RF-BOOKING-LOOKUP-04, RF-BOOKING-LOOKUP-AUTH

Resources: none

Examples

```
booking lookup BOOK001
```

Assumptions: none

Sources: SRC-BOOKING-BRIEF

Open issues: none

Appendix: graphs

Identity / name	Contract	Description / details
GRAPH-VLD-BOOKING-CODE	validation	VLD-BOOKING-CODE; Figure
GRAPH-VLD-BOOKING-CANCEL	validation	VLD-BOOKING-CANCEL; Figure
GRAPH-BOOKING-LOOKUP	use_case	UC-BOOKING-LOOKUP; Figure
GRAPH-BOOKING-CREATE	use_case	UC-BOOKING-CREATE; Figure
GRAPH-BOOKING-CANCEL	use_case	UC-BOOKING-CANCEL; Figure
GRAPH-BOOKING-CANCEL-REFUSAL	use_case	UC-BOOKING-CANCEL; Figure
GRAPH-VLD-BOOKING-TITLE	validation	VLD-BOOKING-TITLE; Figure

Graph `GRAPH-VLD-BOOKING-CODE`

Booking code validation

Kind: flow

Nodes

Identity / name	Contract	Description / details
N-VLD-BOOK-CODE-START	Type: start	Label: Start Requirement refs: none
N-VLD-BOOK-CODE-READ	Type: action	Label: Read code Requirement refs: none
N-VLD-BOOK-CODE-EMPTY	Type: decision	Label: Non-empty? Requirement refs: none
N-VLD-BOOK-CODE-LENGTH	Type: decision	Label: 1 to 12 characters? Requirement refs: none
N-VLD-BOOK-CODE-CHARS	Type: decision	Label: Uppercase letters or digits? Requirement refs: none
N-VLD-BOOK-CODE-PASS	Type: result	Label: Valid Requirement refs: none
N-VLD-BOOK-CODE-FAIL	Type: refusal	Label: Invalid code Requirement refs: none

Edges

Identity / name	Contract	Description / details
E-VLD-BOOK-CODE-1	From: N-VLD-BOOK-CODE-START To: N-VLD-BOOK-CODE-READ	Requirement refs: none
E-VLD-BOOK-CODE-2	From: N-VLD-BOOK-CODE-READ To: N-VLD-BOOK-CODE-EMPTY	Requirement refs: none
E-VLD-BOOK-CODE-3	From: N-VLD-BOOK-CODE-EMPTY To: N-VLD-BOOK-CODE-FAIL	Requirement refs: none Label: no
E-VLD-BOOK-CODE-4	From: N-VLD-BOOK-CODE-EMPTY To: N-VLD-BOOK-CODE-LENGTH	Requirement refs: none Label: yes
E-VLD-BOOK-CODE-5	From: N-VLD-BOOK-CODE-LENGTH To: N-VLD-BOOK-CODE-FAIL	Requirement refs: none Label: no

Identity / name	Contract	Description / details
E-VLD-BOOK-CODE-6	From: N-VLD-BOOK-CODE-LENGTH To: N-VLD-BOOK-CODE-CHARS	Requirement refs: none Label: yes
E-VLD-BOOK-CODE-7	From: N-VLD-BOOK-CODE-CHARS To: N-VLD-BOOK-CODE-FAIL	Requirement refs: none Label: no
E-VLD-BOOK-CODE-8	From: N-VLD-BOOK-CODE-CHARS To: N-VLD-BOOK-CODE-PASS	Requirement refs: none Label: yes

Topology of GRAPH-VLD-BOOKING-CODE:
Nodes:
- N-VLD-BOOK-CODE-START (start)
- N-VLD-BOOK-CODE-READ (action)
- N-VLD-BOOK-CODE-EMPTY (decision)
- N-VLD-BOOK-CODE-LENGTH (decision)
- N-VLD-BOOK-CODE-CHARS (decision)
- N-VLD-BOOK-CODE-PASS (result)
- N-VLD-BOOK-CODE-FAIL (refusal)

Edges:
- E-VLD-BOOK-CODE-1: N-VLD-BOOK-CODE-START -> N-VLD-BOOK-CODE-READ
- E-VLD-BOOK-CODE-2: N-VLD-BOOK-CODE-READ -> N-VLD-BOOK-CODE-EMPTY
- E-VLD-BOOK-CODE-3: N-VLD-BOOK-CODE-EMPTY -> N-VLD-BOOK-CODE-FAIL
- E-VLD-BOOK-CODE-4: N-VLD-BOOK-CODE-EMPTY -> N-VLD-BOOK-CODE-LENGTH
- E-VLD-BOOK-CODE-5: N-VLD-BOOK-CODE-LENGTH -> N-VLD-BOOK-CODE-FAIL
- E-VLD-BOOK-CODE-6: N-VLD-BOOK-CODE-LENGTH -> N-VLD-BOOK-CODE-CHARS
- E-VLD-BOOK-CODE-7: N-VLD-BOOK-CODE-CHARS -> N-VLD-BOOK-CODE-FAIL
- E-VLD-BOOK-CODE-8: N-VLD-BOOK-CODE-CHARS -> N-VLD-BOOK-CODE-PASS

Figure: GRAPH-VLD-BOOKING-CODE

Graph `GRAPH-VLD-BOOKING-CANCEL`

Cancellation transition validation

Kind: flow

Nodes

Identity / name	Contract	Description / details
N-VLD-BOOK-CANCEL-START	Type: start	Label: Start Requirement refs: none
N-VLD-BOOK-CANCEL-READ	Type: action	Label: Read booking status Requirement refs: none
N-VLD-BOOK-CANCEL-ACTIVE	Type: decision	Label: Status active? Requirement refs: none
N-VLD-BOOK-CANCEL-PASS	Type: result	Label: Transition allowed Requirement refs: none

Identity / name	Contract	Description / details
N-VLD-BOOK-CANCEL-FAIL	Type: refusal	Label: Already cancelled Requirement refs: none

Edges

Identity / name	Contract	Description / details
E-VLD-BOOK-CANCEL-1	From: N-VLD-BOOK-CANCEL-START To: N-VLD-BOOK-CANCEL-READ	Requirement refs: none
E-VLD-BOOK-CANCEL-2	From: N-VLD-BOOK-CANCEL-READ To: N-VLD-BOOK-CANCEL-ACTIVE	Requirement refs: none
E-VLD-BOOK-CANCEL-3	From: N-VLD-BOOK-CANCEL-ACTIVE To: N-VLD-BOOK-CANCEL-FAIL	Requirement refs: none Label: no
E-VLD-BOOK-CANCEL-4	From: N-VLD-BOOK-CANCEL-ACTIVE To: N-VLD-BOOK-CANCEL-PASS	Requirement refs: none Label: yes

Topology of GRAPH-VLD-BOOKING-CANCEL:
Nodes:
- N-VLD-BOOK-CANCEL-START (start)
- N-VLD-BOOK-CANCEL-READ (action)
- N-VLD-BOOK-CANCEL-ACTIVE (decision)
- N-VLD-BOOK-CANCEL-PASS (result)
- N-VLD-BOOK-CANCEL-FAIL (refusal)

Edges:
- E-VLD-BOOK-CANCEL-1: N-VLD-BOOK-CANCEL-START -> N-VLD-BOOK-CANCEL-READ
- E-VLD-BOOK-CANCEL-2: N-VLD-BOOK-CANCEL-READ -> N-VLD-BOOK-CANCEL-ACTIVE
- E-VLD-BOOK-CANCEL-3: N-VLD-BOOK-CANCEL-ACTIVE -> N-VLD-BOOK-CANCEL-FAIL
- E-VLD-BOOK-CANCEL-4: N-VLD-BOOK-CANCEL-ACTIVE -> N-VLD-BOOK-CANCEL-PASS

Figure: GRAPH-VLD-BOOKING-CANCEL

Graph `GRAPH-BOOKING-LOOKUP`

Locate booking

Kind: flow

Nodes

Identity / name	Contract	Description / details
N-BOOK-LOOK-START	Type: start	Label: Start Requirement refs: none
N-BOOKING-LOOKUP-AUTH	Type: decision	Label: Trusted permission? Requirement refs: RF-BOOKING-LOOKUP-AUTH
N-BOOKING-LOOKUP-DENIED	Type: refusal	Label: Denied; no collaborator Requirement refs: RF-BOOKING-LOOKUP-AUTH
N-BOOK-LOOK-CODE	Type: validation	Label: Apply VLD-BOOKING-CODE Requirement refs: RF-BOOKING-LOOKUP-01 Call: Kind: validation; Identifier: VLD-BOOKING-CODE
N-BOOK-LOOK-CODEC	Type: decision	Label: Code valid? Requirement refs: RF-BOOKING-LOOKUP-01
N-BOOK-LOOK-INVALID	Type: refusal	Label: Refuse: invalid code Requirement refs: RF-BOOKING-LOOKUP-01
N-BOOK-LOOK-LOAD	Type: operation	Label: Load booking Requirement refs: RF-BOOKING-LOOKUP-02 Call: Kind: operation; Identifier: OP-BOOKING-LOAD
N-BOOK-LOOK-NOTFOUND	Type: refusal	Label: Refuse: not found Requirement refs: RF-BOOKING-LOOKUP-03
N-BOOK-LOOK-FAIL	Type: failure	Label: Report store unavailable Requirement refs: RF-BOOKING-LOOKUP-04
N-BOOK-LOOK-FOUND	Type: result	Label: Booking found Requirement refs: RF-BOOKING-LOOKUP-02

Edges

Identity / name	Contract	Description / details
E-BOOK-LOOK-1	From: N-BOOK-LOOK-START To: N-BOOKING-LOOKUP-AUTH	Requirement refs: none
E-BOOK-LOOK-2	From: N-BOOK-LOOK-CODE To: N-BOOK-LOOK-CODEC	Requirement refs: none
E-BOOK-LOOK-3	From: N-BOOK-LOOK-CODEC To: N-BOOK-LOOK-INVALID	Requirement refs: RF-BOOKING-LOOKUP-01 Label: fail
E-BOOK-LOOK-4	From: N-BOOK-LOOK-CODEC To: N-BOOK-LOOK-LOAD	Requirement refs: RF-BOOKING-LOOKUP-02 Label: pass

Identity / name	Contract	Description / details
E-BOOK-LOOK-5	From: N-BOOK-LOOK-LOAD To: N-BOOK-LOOK-NOTFOUND	Requirement refs: RF-BOOKING-LOOKUP-03 Label: NOT_FOUND
E-BOOK-LOOK-6	From: N-BOOK-LOOK-LOAD To: N-BOOK-LOOK-FAIL	Requirement refs: RF-BOOKING-LOOKUP-04 Label: STORE_UNAVAILABLE
E-BOOK-LOOK-7	From: N-BOOK-LOOK-LOAD To: N-BOOK-LOOK-FOUND	Requirement refs: RF-BOOKING-LOOKUP-02 Label: loaded
E-BOOKING-LOOKUP-ALLOW	From: N-BOOKING-LOOKUP-AUTH To: N-BOOK-LOOK-CODE	Requirement refs: RF-BOOKING-LOOKUP-AUTH Label: authorized
E-BOOKING-LOOKUP-DENY	From: N-BOOKING-LOOKUP-AUTH To: N-BOOKING-LOOKUP-DENIED	Requirement refs: RF-BOOKING-LOOKUP-AUTH Label: missing / unresolved / insufficient

Topology of GRAPH-BOOKING-LOOKUP:
Nodes:
- N-BOOK-LOOK-START (start)
- N-BOOKING-LOOKUP-AUTH (decision)
- N-BOOKING-LOOKUP-DENIED (refusal)
- N-BOOK-LOOK-CODE (validation)
- N-BOOK-LOOK-CODEC (decision)
- N-BOOK-LOOK-INVALID (refusal)
- N-BOOK-LOOK-LOAD (operation)
- N-BOOK-LOOK-NOTFOUND (refusal)
- N-BOOK-LOOK-FAIL (failure)
- N-BOOK-LOOK-FOUND (result)

Edges:
- E-BOOK-LOOK-1: N-BOOK-LOOK-START -> N-BOOKING-LOOKUP-AUTH
- E-BOOK-LOOK-2: N-BOOK-LOOK-CODE -> N-BOOK-LOOK-CODEC
- E-BOOK-LOOK-3: N-BOOK-LOOK-CODEC -> N-BOOK-LOOK-INVALID
- E-BOOK-LOOK-4: N-BOOK-LOOK-CODEC -> N-BOOK-LOOK-LOAD
- E-BOOK-LOOK-5: N-BOOK-LOOK-LOAD -> N-BOOK-LOOK-NOTFOUND
- E-BOOK-LOOK-6: N-BOOK-LOOK-LOAD -> N-BOOK-LOOK-FAIL
- E-BOOK-LOOK-7: N-BOOK-LOOK-LOAD -> N-BOOK-LOOK-FOUND
- E-BOOKING-LOOKUP-ALLOW: N-BOOKING-LOOKUP-AUTH -> N-BOOK-LOOK-CODE
- E-BOOKING-LOOKUP-DENY: N-BOOKING-LOOKUP-AUTH -> N-BOOKING-LOOKUP-DENIED

Figure: GRAPH-BOOKING-LOOKUP

Graph `GRAPH-BOOKING-CREATE`

Create booking

Kind: flow

Nodes

Identity / name	Contract	Description / details
N-BOOK-CREATE-START	Type: start	Label: Start Requirement refs: none
N-BOOKING-CREATE-AUTH	Type: decision	Label: Trusted permission? Requirement refs: RF-BOOKING-CREATE-AUTH
N-BOOKING-CREATE-DENIED	Type: refusal	Label: Denied; no collaborator Requirement refs: RF-BOOKING-CREATE-AUTH
N-BOOK-CREATE-CODE	Type: validation	Label: Apply VLD-BOOKING-CODE Requirement refs: RF-BOOKING-CREATE-01 Call: Kind: validation; Identifier: VLD-BOOKING-CODE
N-BOOK-CREATE-CODEC	Type: decision	Label: Code valid? Requirement refs: RF-BOOKING-CREATE-01
N-BOOK-CREATE-INVALID	Type: refusal	Label: Invalid code or title Requirement refs: RF-BOOKING-CREATE-01
N-BOOK-CREATE-SAVE	Type: operation	Label: Save active booking Requirement refs: RF-BOOKING-CREATE-02 Call: Kind: operation; Identifier: OP-BOOKING-SAVE
N-BOOK-CREATE-DUP	Type: failure	Label: Report VERSION_CONFLICT Requirement refs: RF-BOOKING-CREATE-03
N-BOOK-CREATE-FAIL	Type: failure	Label: Report STORE_UNAVAILABLE Requirement refs: RF-BOOKING-CREATE-03
N-BOOK-CREATE-DONE	Type: result	Label: Booking created Requirement refs: RF-BOOKING-CREATE-04
N-BOOK-CREATE-TITLE	Type: validation	Label: Apply title validity Requirement refs: RF-BOOKING-CREATE-TITLE Call: Kind: validation; Identifier: VLD-BOOKING-TITLE
N-BOOK-CREATE-TITLEVALID	Type: decision	Label: Title valid? Requirement refs: RF-BOOKING-CREATE-TITLE

Edges

Identity / name	Contract	Description / details
E-BOOK-CREATE-1	From: N-BOOK-CREATE-START To: N-BOOKING-CREATE-AUTH	Requirement refs: none

Identity / name	Contract	Description / details
E-BOOK-CREATE-2	From: N-BOOK-CREATE-CODE To: N-BOOK-CREATE-CODEC	Requirement refs: none
E-BOOK-CREATE-3	From: N-BOOK-CREATE-CODEC To: N-BOOK-CREATE-INVALID	Requirement refs: RF-BOOKING-CREATE-01 Label: fail
E-BOOK-CREATE-4	From: N-BOOK-CREATE-CODEC To: N-BOOK-CREATE-TITLE	Requirement refs: RF-BOOKING-CREATE-02 Label: pass
E-BOOK-CREATE-5	From: N-BOOK-CREATE-SAVE To: N-BOOK-CREATE-DUP	Requirement refs: RF-BOOKING-CREATE-03 Label: VERSION_CONFLICT
E-BOOK-CREATE-6	From: N-BOOK-CREATE-SAVE To: N-BOOK-CREATE-FAIL	Requirement refs: RF-BOOKING-CREATE-03 Label: STORE_UNAVAILABLE
E-BOOK-CREATE-7	From: N-BOOK-CREATE-SAVE To: N-BOOK-CREATE-DONE	Requirement refs: RF-BOOKING-CREATE-02; RF-BOOKING-CREATE-04 Label: saved
E-BOOK-CREATE-TITLE-1	From: N-BOOK-CREATE-TITLE To: N-BOOK-CREATE-TITLEVALID	Requirement refs: none
E-BOOK-CREATE-TITLE-2	From: N-BOOK-CREATE-TITLEVALID To: N-BOOK-CREATE-INVALID	Requirement refs: RF-BOOKING-CREATE-TITLE Label: fail
E-BOOK-CREATE-TITLE-3	From: N-BOOK-CREATE-TITLEVALID To: N-BOOK-CREATE-SAVE	Requirement refs: RF-BOOKING-CREATE-TITLE Label: pass
E-BOOKING-CREATE-ALLOW	From: N-BOOKING-CREATE-AUTH To: N-BOOK-CREATE-CODE	Requirement refs: RF-BOOKING-CREATE-AUTH Label: authorized
E-BOOKING-CREATE-DENY	From: N-BOOKING-CREATE-AUTH To: N-BOOKING-CREATE-DENIED	Requirement refs: RF-BOOKING-CREATE-AUTH Label: missing / unresolved / insufficient

Topology of GRAPH-BOOKING-CREATE:
Nodes:
- N-BOOK-CREATE-START (start)

- N-BOOKING-CREATE-AUTH (decision)
- N-BOOKING-CREATE-DENIED (refusal)
- N-BOOK-CREATE-CODE (validation)
- N-BOOK-CREATE-CODEC (decision)
- N-BOOK-CREATE-INVALID (refusal)
- N-BOOK-CREATE-SAVE (operation)
- N-BOOK-CREATE-DUP (failure)
- N-BOOK-CREATE-FAIL (failure)
- N-BOOK-CREATE-DONE (result)
- N-BOOK-CREATE-TITLE (validation)
- N-BOOK-CREATE-TITLEVALID (decision)

Edges:
- E-BOOK-CREATE-1: N-BOOK-CREATE-START -> N-BOOKING-CREATE-AUTH
- E-BOOK-CREATE-2: N-BOOK-CREATE-CODE -> N-BOOK-CREATE-CODEC
- E-BOOK-CREATE-3: N-BOOK-CREATE-CODEC -> N-BOOK-CREATE-INVALID
- E-BOOK-CREATE-4: N-BOOK-CREATE-CODEC -> N-BOOK-CREATE-TITLE
- E-BOOK-CREATE-5: N-BOOK-CREATE-SAVE -> N-BOOK-CREATE-DUP
- E-BOOK-CREATE-6: N-BOOK-CREATE-SAVE -> N-BOOK-CREATE-FAIL
- E-BOOK-CREATE-7: N-BOOK-CREATE-SAVE -> N-BOOK-CREATE-DONE
- E-BOOK-CREATE-TITLE-1: N-BOOK-CREATE-TITLE -> N-BOOK-CREATE-TITLEVALID
- E-BOOK-CREATE-TITLE-2: N-BOOK-CREATE-TITLEVALID -> N-BOOK-CREATE-INVALID
- E-BOOK-CREATE-TITLE-3: N-BOOK-CREATE-TITLEVALID -> N-BOOK-CREATE-SAVE
- E-BOOKING-CREATE-ALLOW: N-BOOKING-CREATE-AUTH -> N-BOOK-CREATE-CODE
- E-BOOKING-CREATE-DENY: N-BOOKING-CREATE-AUTH -> N-BOOKING-CREATE-DENIED

Figure: GRAPH-BOOKING-CREATE

Graph `GRAPH-BOOKING-CANCEL`

Cancel booking

Kind: flow

Nodes

Identity / name	Contract	Description / details
N-BOOK-CANCEL-START	Type: start	Label: Start Requirement refs: none
N-BOOKING-CANCEL-AUTH	Type: decision	Label: Trusted permission? Requirement refs: RF-BOOKING-CANCEL-AUTH
N-BOOKING-CANCEL-DENIED	Type: refusal	Label: Denied; no collaborator Requirement refs: RF-BOOKING-CANCEL-AUTH
N-BOOK-CANCEL-LOOK	Type: use_case	Label: Call UC-BOOKING-LOOKUP Requirement refs: RF-BOOKING-CANCEL-01 Call: Kind: use_case; Identifier: UC-BOOKING-LOOKUP
N-BOOK-CANCEL-FOUND	Type: decision	Label: Booking found? Requirement refs: RF-BOOKING-CANCEL-01
N-BOOK-CANCEL-NOTFOUND	Type: refusal	Label: Refuse: not found Requirement refs: RF-BOOKING-CANCEL-01

Identity / name	Contract	Description / details
N-BOOK-CANCEL-VALIDATE	Type: validation	Label: Apply VLD-BOOKING-CANCEL Requirement refs: RF-BOOKING-CANCEL-02 Call: Kind: validation; Identifier: VLD-BOOKING-CANCEL
N-BOOK-CANCEL-VALID	Type: decision	Label: Transition allowed? Requirement refs: RF-BOOKING-CANCEL-02
N-BOOK-CANCEL-ALREADY	Type: refusal	Label: Refuse: already cancelled Requirement refs: RF-BOOKING-CANCEL-03
N-BOOK-CANCEL-SAVE	Type: operation	Label: Save cancelled booking Requirement refs: RF-BOOKING-CANCEL-04 Call: Kind: operation; Identifier: OP-BOOKING-SAVE
N-BOOK-CANCEL-ERROR	Type: failure	Label: Report store failure Requirement refs: RF-BOOKING-CANCEL-05
N-BOOK-CANCEL-DONE	Type: result	Label: Booking cancelled Requirement refs: RF-BOOKING-CANCEL-04
N-BOOK-CANCEL-INVALID	Type: refusal	Label: Invalid code; no write Requirement refs: RF-BOOKING-CANCEL-01
N-BOOK-CANCEL-LOOKDENIED	Type: refusal	Label: Lookup denied; no write Requirement refs: RF-BOOKING-CANCEL-01

Edges

Identity / name	Contract	Description / details
E-BOOK-CANCEL-1	From: N-BOOK-CANCEL-START To: N-BOOKING-CANCEL-AUTH	Requirement refs: none
E-BOOK-CANCEL-2	From: N-BOOK-CANCEL-LOOK To: N-BOOK-CANCEL-FOUND	Requirement refs: none Label: found / not found
E-BOOK-CANCEL-3	From: N-BOOK-CANCEL-FOUND To: N-BOOK-CANCEL-NOTFOUND	Requirement refs: RF-BOOKING-CANCEL-01 Label: no
E-BOOK-CANCEL-4	From: N-BOOK-CANCEL-FOUND To: N-BOOK-CANCEL-VALIDATE	Requirement refs: RF-BOOKING-CANCEL-01 Label: yes
E-BOOK-CANCEL-5	From: N-BOOK-CANCEL-VALIDATE To: N-BOOK-CANCEL-VALID	Requirement refs: RF-BOOKING-CANCEL-02

Identity / name	Contract	Description / details
E-BOOK-CANCEL-6	From: N-BOOK-CANCEL-VALID To: N-BOOK-CANCEL-ALREADY	Requirement refs: RF-BOOKING-CANCEL-03 Label: fail
E-BOOK-CANCEL-7	From: N-BOOK-CANCEL-VALID To: N-BOOK-CANCEL-SAVE	Requirement refs: RF-BOOKING-CANCEL-04 Label: pass
E-BOOK-CANCEL-8	From: N-BOOK-CANCEL-SAVE To: N-BOOK-CANCEL-ERROR	Requirement refs: RF-BOOKING-CANCEL-05 Label: failure
E-BOOK-CANCEL-9	From: N-BOOK-CANCEL-SAVE To: N-BOOK-CANCEL-DONE	Requirement refs: RF-BOOKING-CANCEL-04 Label: saved
E-BOOK-CANCEL-INVALID	From: N-BOOK-CANCEL-LOOK To: N-BOOK-CANCEL-INVALID	Requirement refs: RF-BOOKING-CANCEL-01 Label: invalid code
E-BOOK-CANCEL-LOOKERROR	From: N-BOOK-CANCEL-LOOK To: N-BOOK-CANCEL-ERROR	Requirement refs: RF-BOOKING-CANCEL-05 Label: lookup failure
E-BOOK-CANCEL-LOOKDENIED	From: N-BOOK-CANCEL-LOOK To: N-BOOK-CANCEL-LOOKDENIED	Requirement refs: RF-BOOKING-CANCEL-01 Label: lookup denied
E-BOOKING-CANCEL-ALLOW	From: N-BOOKING-CANCEL-AUTH To: N-BOOK-CANCEL-LOOK	Requirement refs: RF-BOOKING-CANCEL-AUTH Label: authorized
E-BOOKING-CANCEL-DENY	From: N-BOOKING-CANCEL-AUTH To: N-BOOKING-CANCEL-DENIED	Requirement refs: RF-BOOKING-CANCEL-AUTH Label: missing / unresolved / insufficient

Topology of GRAPH-BOOKING-CANCEL:
Nodes:
- N-BOOK-CANCEL-START (start)
- N-BOOKING-CANCEL-AUTH (decision)
- N-BOOKING-CANCEL-DENIED (refusal)
- N-BOOK-CANCEL-LOOK (use_case)
- N-BOOK-CANCEL-FOUND (decision)
- N-BOOK-CANCEL-NOTFOUND (refusal)
- N-BOOK-CANCEL-VALIDATE (validation)
- N-BOOK-CANCEL-VALID (decision)
- N-BOOK-CANCEL-ALREADY (refusal)
- N-BOOK-CANCEL-SAVE (operation)
- N-BOOK-CANCEL-ERROR (failure)
- N-BOOK-CANCEL-DONE (result)

- N-BOOK-CANCEL-INVALID (refusal)
- N-BOOK-CANCEL-LOOKDENIED (refusal)
Edges:
- E-BOOK-CANCEL-1: N-BOOK-CANCEL-START -> N-BOOKING-CANCEL-AUTH
- E-BOOK-CANCEL-2: N-BOOK-CANCEL-LOOK -> N-BOOK-CANCEL-FOUND
- E-BOOK-CANCEL-3: N-BOOK-CANCEL-FOUND -> N-BOOK-CANCEL-NOTFOUND
- E-BOOK-CANCEL-4: N-BOOK-CANCEL-FOUND -> N-BOOK-CANCEL-VALIDATE
- E-BOOK-CANCEL-5: N-BOOK-CANCEL-VALIDATE -> N-BOOK-CANCEL-VALID
- E-BOOK-CANCEL-6: N-BOOK-CANCEL-VALID -> N-BOOK-CANCEL-ALREADY
- E-BOOK-CANCEL-7: N-BOOK-CANCEL-VALID -> N-BOOK-CANCEL-SAVE
- E-BOOK-CANCEL-8: N-BOOK-CANCEL-SAVE -> N-BOOK-CANCEL-ERROR
- E-BOOK-CANCEL-9: N-BOOK-CANCEL-SAVE -> N-BOOK-CANCEL-DONE
- E-BOOK-CANCEL-INVALID: N-BOOK-CANCEL-LOOK -> N-BOOK-CANCEL-INVALID
- E-BOOK-CANCEL-LOOKERROR: N-BOOK-CANCEL-LOOK -> N-BOOK-CANCEL-ERROR
- E-BOOK-CANCEL-LOOKDENIED: N-BOOK-CANCEL-LOOK -> N-BOOK-CANCEL-LOOKDENIED
- E-BOOKING-CANCEL-ALLOW: N-BOOKING-CANCEL-AUTH -> N-BOOK-CANCEL-LOOK
- E-BOOKING-CANCEL-DENY: N-BOOKING-CANCEL-AUTH -> N-BOOKING-CANCEL-DENIED

Figure: GRAPH-BOOKING-CANCEL

Graph `GRAPH-BOOKING-CANCEL-REFUSAL`

Cancellation refusal subflow

Kind: flow

Nodes

Identity / name	Contract	Description / details
N-BOOK-REFUSE-START	Type: start	Label: Refusal Requirement refs: RF-BOOKING-CANCEL-03
N-BOOK-REFUSE-ALREADY	Type: refusal	Label: Already cancelled: no effect Requirement refs: RF-BOOKING-CANCEL-03
N-BOOK-REFUSE-NOTFOUND	Type: refusal	Label: Not found: no effect Requirement refs: RF-BOOKING-CANCEL-01

Edges

Identity / name	Contract	Description / details
E-BOOK-REFUSE-1	From: N-BOOK-REFUSE-START To: N-BOOK-REFUSE-ALREADY	Requirement refs: RF-BOOKING-CANCEL-03 Label: already cancelled
E-BOOK-REFUSE-2	From: N-BOOK-REFUSE-START To: N-BOOK-REFUSE-NOTFOUND	Requirement refs: RF-BOOKING-CANCEL-01 Label: not found

Topology of GRAPH-BOOKING-CANCEL-REFUSAL:
Nodes:
- N-BOOK-REFUSE-START (start)
- N-BOOK-REFUSE-ALREADY (refusal)
- N-BOOK-REFUSE-NOTFOUND (refusal)
Edges:

- E-BOOK-REFUSE-1: N-BOOK-REFUSE-START -> N-BOOK-REFUSE-ALREADY
- E-BOOK-REFUSE-2: N-BOOK-REFUSE-START -> N-BOOK-REFUSE-NOTFOUND

Figure: GRAPH-BOOKING-CANCEL-REFUSAL

Graph `GRAPH-VLD-BOOKING-TITLE`

Booking title validity

Kind: flow

Nodes

Identity / name	Contract	Description / details
N-VLD-BOOK-TITLE-START	Type: start	Label: Receive title Requirement refs: none
N-VLD-BOOK-TITLE-CHECK	Type: decision	Label: Meaningful text? Requirement refs: none
N-VLD-BOOK-TITLE-PASS	Type: result	Label: Pass unchanged Requirement refs: none
N-VLD-BOOK-TITLE-FAIL	Type: refusal	Label: Invalid title Requirement refs: none

Edges

Identity / name	Contract	Description / details
E-VLD-BOOK-TITLE-1	From: N-VLD-BOOK-TITLE-START To: N-VLD-BOOK-TITLE-CHECK	Requirement refs: none
E-VLD-BOOK-TITLE-2	From: N-VLD-BOOK-TITLE-CHECK To: N-VLD-BOOK-TITLE-PASS	Requirement refs: none Label: yes
E-VLD-BOOK-TITLE-3	From: N-VLD-BOOK-TITLE-CHECK To: N-VLD-BOOK-TITLE-FAIL	Requirement refs: none Label: no

Topology of GRAPH-VLD-BOOKING-TITLE:
Nodes:
- N-VLD-BOOK-TITLE-START (start)
- N-VLD-BOOK-TITLE-CHECK (decision)
- N-VLD-BOOK-TITLE-PASS (result)
- N-VLD-BOOK-TITLE-FAIL (refusal)

Edges:
- E-VLD-BOOK-TITLE-1: N-VLD-BOOK-TITLE-START -> N-VLD-BOOK-TITLE-CHECK
- E-VLD-BOOK-TITLE-2: N-VLD-BOOK-TITLE-CHECK -> N-VLD-BOOK-TITLE-PASS
- E-VLD-BOOK-TITLE-3: N-VLD-BOOK-TITLE-CHECK -> N-VLD-BOOK-TITLE-FAIL

Figure: GRAPH-VLD-BOOKING-TITLE

Appendix: sources

SRC-BOOKING-BRIEF

Title	Synthetic booking-desk teaching brief	Uri	null
Revision	2	Accessed	null

Note

Synthetic example authored for Atlas. No stakeholder interview, independent approval, or accountable human Needs-to-L1 derivation is asserted.

The authorization policy, interaction sketches, and quality targets are didactic choices, not observed business facts.

Appendix: history

HIST-BOOKING-01

Version	1	Date	2026-10-25

Description

Initial reference example for the first Atlas Specifications edition, Alpha-v0.1.0. This synthetic example demonstrates the model and its representations; it does not assert stakeholder approval or execution of the described business application.

HIST-MIGRATION-STABLE

Version	2	Date	2026-10-25

Description

Explicit editorial migration from Atlas Alpha-v0.1.0 revision 1 to Atlas Stable-v1.0.0 revision 2. Model/profile/envelope and producer identities change; existing concept IDs, domain contracts, authored history and resources are retained. This is not format conversion, a Needs-to-L1 derivation, or new stakeholder approval.

Derived summary: 5 sections; 1 actors; 1 entities; 1 tools; 3 use cases; 1 boundaries; 7 graphs; 0 resources.